BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Chemistry — SSC-I (9th Grade)

Syllabus: Chapter 14 — Hydrocarbons

Total Marks: 65	**Time Allowed: 2h 30m**	**Version: 1**

General Instructions:
1. Write your name, roll number and section clearly on the answer sheet.
2. Section A is compulsory. Encircle the correct option on the question paper itself.
3. In Section B, attempt **all 11** questions. For each question, attempt either the main question **OR** its alternative.
4. In Section C, attempt **all 4** questions. For each question, attempt either the main question **OR** its alternative.
5. Use black or blue ink only. Pencil is not allowed except for diagrams.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	A	B	C	D	Answer
1	Alkanes are called *saturated* hydrocarbons because they contain only:	C=C double bonds	C≡C triple bonds	C—C single bonds	C=O bonds	ⒶⒷⒸⒹ
2	The general formula of alkanes is:	C_nH_{2n}	C_nH_{2n-2}	C_nH_{2n+2}	C_nH_n	ⒶⒷⒸⒹ
3	The simplest possible alkane molecule is:	Ethane	Propane	Butane	Methane	ⒶⒷⒸⒹ
4	Addition of hydrogen across a C=C double bond is called:	Halogenation	Hydrogenation	Combustion	Cracking	ⒶⒷⒸⒹ
5	Hydrogenation of alkenes and alkynes requires a catalyst. Which catalyst is used at 200–300°C?	Iron (Fe)	Copper (Cu)	Nickel (Ni)	Platinum (Pt)	ⒶⒷⒸⒹ
6	In the reduction of alkyl halides, the reagent used to produce nascent hydrogen is:	Zn/HCl(aq)	Ni, high pressure	Sunlight	Pt/Pd at room temp	ⒶⒷⒸⒹ
7	Breaking a large hydrocarbon into smaller ones by heating at 450–750°C and high pressure is called:	Combustion	Hydrogenation	Cracking	Halogenation	ⒶⒷⒸⒹ
8	When decane ($C_{10}H_{22}$) undergoes cracking, the products are:	Octane + Ethene	Hexane + Butane	Propane + Heptane	Methane + Nonane	ⒶⒷⒸⒹ
9	Alkanes react with halogens in sunlight via *substitution*. This means one hydrogen atom is:	Added alongside existing H	Replaced by a halogen atom	Removed without replacement	Oxidized to water	ⒶⒷⒸⒹ
10	$CH_4 + Cl_2$ in diffused sunlight gives:	$CH_3Cl + HCl$	$CH_2Cl_2 + H_2$	$CCl_4 + 2H_2$	$CHCl_3 + HCl$	ⒶⒷⒸⒹ
11	Complete combustion of methane produces:	$CO + H_2O$	$CO_2 + H_2O$ + heat	$CO_2 + C + H_2O$	$C + HCl$	ⒶⒷⒸⒹ
12	Alkanes are generally *unreactive* towards ionic compounds because their molecules are:	polar	non-polar	ionic	metallic	ⒶⒷⒸⒹ

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt all 11 questions. For each pair, attempt either the main question OR its alternative.

Part	Question	Marks	OR	OR Question	Marks
2(i)	Define hydrocarbons. How are they classified according to the type of bond between carbon atoms? Give one example of each class.	**3**	**OR**	Define alkanes. State their general formula. Give the molecular formulae and names of the first two members of the alkane series.	**3**
2(ii)	Draw the structural formula of methane (CH_4) and ethane (C_2H_6). What type of bond is present in each?	**3**	**OR**	Draw the structural formula of an alkane containing five carbon atoms. Write its molecular formula.	**3**
2(iii)	What is hydrogenation? Write the equation for the hydrogenation of ethene to produce ethane. State the catalyst and temperature used.	**3**	**OR**	Write the equation for the hydrogenation of ethyne (CH≡CH) to produce ethene. Why does an alkyne require two molecules of hydrogen for complete reduction to an alkane?	**3**
2(iv)	Describe the reduction of alkyl halides to prepare an alkane. Write the equation for the reduction of chloromethane (CH_3Cl) and state what nascent hydrogen is.	**3**	**OR**	Complete the following reaction and name the product: $CH_3—CH_2—Cl + 2[H] \rightarrow ?$ State the reagent providing 2[H] and name	**3**

				the type of reaction.	
2(v)	What is cracking? Write the equation showing cracking of decane ($C_{10}H_{22}$). Name the two products and state the conditions required.	**3**	**OR**	Explain why cracking is an important industrial process. What types of products are formed during thermal cracking?	**3**
2(vi)	Describe two physical properties of alkanes: (a) polarity and solubility in water; (b) states at room temperature based on carbon chain length.	**3**	**OR**	Hexane is used as a solvent to extract vegetable oils. Explain why alkanes are suitable for this purpose based on their chemical and physical properties.	**3**
2(vii)	Define substitution reaction. Write the equation for the reaction of methane with chlorine in diffused sunlight. Name the organic product.	**3**	**OR**	What happens when methane reacts with chlorine in *direct* sunlight instead of diffused sunlight? Write the equation and describe the difference in products.	**3**
2(viii)	Why is the reaction of alkanes with halogens called a *photochemical* reaction? What does this term mean?	**3**	**OR**	Alkanes do not undergo addition reactions but undergo substitution reactions. Explain why, linking your answer to the type of bonding in alkanes.	**3**
2(ix)	Define combustion. Write the equation for the complete combustion of methane. Name all products and state the type of flame produced.	**3**	**OR**	Write the equation for the incomplete combustion of methane ($3CH_4 + 4O_2$). Name all three products and explain when incomplete combustion occurs.	**3**
2(x)	Give three reasons why lighter alkanes are widely used as fuels.	**3**	**OR**	Methane is the main component of natural gas. State three properties of methane that make it useful as a domestic fuel.	**3**
2(xi)	Write the chemical equations for the preparation of propane (C_3H_8) by: (a) hydrogenation of propene; (b) reduction of 1-chloropropane ($CH_3CH_2CH_2Cl$).	**3**	**OR**	Using equations, show how ethane (C_2H_6) can be prepared by (a) cracking and (b) hydrogenation. Include all conditions and reagents.	**3**

SECTION C — Long Questions (4 × 5 = 20 Marks) — Attempt all 4

Part	Question	Marks	OR	OR Question	Marks
3(i)	(a) Define alkanes. State their general formula and explain why they are called saturated hydrocarbons. Draw the structural formulas of methane and ethane. [3] (b) Name the TWO chemical reactions alkanes undergo and state the condition required for each. [2]	**5**	**OR**	(a) Describe the three methods of preparing alkanes: hydrogenation, reduction of alkyl halides, cracking. For each, state reagents and conditions. [3] (b) Write one balanced equation illustrating each method. [2]	**5**
3(ii)	(a) Explain hydrogenation. Write balanced equations for: (i) ethyne to ethene; (ii) ethene to ethane. State catalyst and temperature for both. [3] (b) Explain reduction of alkyl halides. Write the equation for reduction of chloromethane. What is nascent hydrogen and how is it produced from Zn/HCl(aq)? [2]	**5**	**OR**	(a) Describe thermal cracking. State the temperature and pressure. Write the equation for cracking of decane into octane and ethene. [3] (b) State two industrial reasons why cracking is important. What mixture of compounds does it typically produce? [2]	**5**
3(iii)	(a) Describe halogenation of alkanes as a photochemical substitution. Define substitution reaction. Write the equation for mono-substitution of methane with Cl_2 in diffused sunlight. [3] (b) Write the equation for the reaction of methane with Cl_2 in direct sunlight. Explain the difference from mono-substitution. [2]	**5**	**OR**	(a) Describe the physical properties of alkanes: polarity, density vs water, solubility in water, and states of matter based on chain length. [3] (b) Explain why alkanes are useful solvents. Name one specific solvent and state what it is used to extract. [2]	**5**
3(iv)	(a) Define combustion. Write balanced equations for complete and incomplete combustion of methane, naming all products in each case. [3] (b) Give three reasons why lighter alkanes make good fuels. Write the equation for combustion of propane (C_3H_8). [2]	**5**	**OR**	(a) Write balanced equations for the preparation of propane (C_3H_8) by: (i) hydrogenation of propene; (ii) reduction of 1-chloropropane; (iii) cracking of a larger alkane of your choice. Include all conditions. [3] (b) Describe photochemical halogenation of ethane with Cl_2. Write the equation for mono-substitution and name the product. [2]	**5**

Invigilator's Signature: ____________________ Candidate's Signature: ____________________

Chapters Covered: Chapter 14 — Hydrocarbons (pp. 193–196)

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